

Part A · Multiple Choice

2 points each

1. **c) 99%**

p. 77 — nitrogen alone is about 78% and oxygen about 21%. Everything else, argon and carbon dioxide included, fits in the remaining 1%.

2. **d) Gravity pulls gas molecules toward Earth's center, and gases near sea level are compressed by the weight of the air above them**

p. 78 — this is also why the force of air weighing down on a unit of area, the atmospheric pressure, is greatest at sea level.

3. **a) troposphere**

p. 80 — because warm air sits beneath cooler air there, the layer is unstable and mixes constantly. That rising and sinking is weather.

4. **b) Stratosphere**

p. 81 — warmer, less dense air sits over cooler, denser air, so nothing overturns. This is why pilots like the lower stratosphere and why volcanic ash can stay suspended there for years.

5. **b) 15 to 30 km**

p. 81 — about 9 to 19 miles. Its thickness varies with both season and latitude.

6. **a) 6.5°C cooler per 1,000 m**

p. 80 — equivalent to 3.6°F per 1,000 feet. A change in temperature with distance is called a temperature gradient.

Part B · True or False

1 point each

7. **TRUE**

p. 80 — inversions can last for days or weeks. They form overland at night or in winter when cold ground chills the air above it, and near coasts where cold seawater does the same.

8. **FALSE**

Density changes greatly with altitude, but composition stays essentially the same. The one exception is the ozone layer, where ozone molecules are far more concentrated. (p. 79)

Part C · Short Answer

3 points each

9. **No UV protection (the ozone layer is in the stratosphere below), extremely low density of gas molecules, and extremely low air pressure.**

Rubric: 1 pt each. **Sunburn** — the ozone layer that blocks UV is in the stratosphere *below* the mesosphere, so there is nothing overhead to filter ultraviolet light. **No oxygen** — 99.9% of the atmosphere's mass lies below the mesosphere, so there is almost nothing left to breathe. **Boiling blood** — air pressure is so low that liquids boil at normal body temperature. (p. 82)

10. Charged particles, trapped by Earth's magnetic field, spiral toward the poles and energize oxygen and nitrogen molecules, making them light up.

Rubric: 1 pt — particles originate in solar flares and travel to Earth on the solar wind, becoming trapped in the Van Allen radiation belts. 1 pt — when a massive solar storm overloads the belts, particles spiral along magnetic field lines toward the poles. 1 pt — the charged particles energize oxygen and nitrogen gas molecules, causing them to emit light, with each gas producing a particular color. (p. 82)